BHUVANESAN N

Software Developer

Java | Spring Boot | MySQL | REST APIs

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GitHub: github.com/Hikki-11

PROFILE SUMMARY

Computer Science undergraduate with hands-on experience in Java, Spring Boot, REST APIs, SQL, and MongoDB, gained through academic and personal software development projects. Strong foundation in Object-Oriented Programming, Data Structures & Algorithms, and backend application development, with 170+ LeetCode problems solved. Experienced in building scalable CRUD applications, integrating databases, and developing IoT-enabled solutions. Passionate about designing efficient backend systems, writing clean, maintainable code, and continuously learning modern software engineering technologies.

PROJECTS

Digital AI Twin IoT ï¿½ Python ï¿½ AI/ML

- ï¿½ Developed a Smart Energy Meter using ESP32 and LCD to monitor and display real-time current, power, and energy consumption.
- ï¿½ Processed sensor inputs to calculate electrical parameters and present live readings on the embedded display.
- ï¿½ Implemented real-time energy monitoring to demonstrate efficient power usage aligned with Sustainable Development Goal (SDG 7).
- ï¿½ Designed an embedded solution capable of continuous energy usage tracking for smart monitoring applications.

Smart Energy Meter ESP32 ï¿½ LCD ï¿½ Embedded Systems ï¿½ IoT

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Employee Management System HTML ï¿½ CSS ï¿½ SQL

- ï¿½ Developed an Employee Management System with a responsive web interface for managing employee records.
- ï¿½ Implemented CRUD operations to add, update, delete, and retrieve employee information using SQL.
- ï¿½ Designed structured input forms for efficient employee data management and validation.
- ï¿½ Built a user-friendly interface to simplify employee record management and improve usability.

Micro Failure Whisperer (Ongoing) Python ï¿½ Machine Learning ï¿½ Predictive Maintenance

- ï¿½ Developing an AI-powered predictive maintenance system to detect micro-level machine failures using sensor and vibration data.
- ï¿½ Applying anomaly detection techniques to identify early signs of equipment degradation before breakdowns occur.
- ï¿½ Designing a real-time monitoring pipeline to process sensor data and generate predictive maintenance alerts.
- ï¿½ Building a scalable architecture capable of supporting industrial machine health monitoring applications.

Digital Smell (Ongoing) IoT ï¿½ Gas Sensors ï¿½ Embedded AI

- ï¿½ Developing an IoT-based gas leakage detection system using high-sensitivity gas sensors and embedded AI.
- ï¿½ Implementing real-time monitoring and anomaly detection to identify minute gas leakages accurately.
- ï¿½ Designing an intelligent alert system to improve industrial safety through early hazard detection.
- ï¿½ Building an embedded solution for continuous environmental monitoring in safety-critical applications.

ScriptGuard AI (Ongoing) Python ï¿½ Computer Vision ï¿½ OCR

- ï¿½ Developing an intelligent paper-scanning mechanism for automated document digitization and verification.
- ï¿½ Integrating OCR and image-processing techniques to extract, validate, and organize scanned content.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C++
Frontend & Web Technologies: HTML5, CSS3, JavaScript, Bootstrap, Responsive Web Design
Backend Technologies: Core Java, Spring, Spring Boot, Spring MVC, REST APIs, JDBC, Hibernate, JPA, CRUD Operations
Databases: MySQL, MongoDB
Tools & Platforms: Git, GitHub, Maven, Postman, IntelliJ IDEA, Eclipse, VS Code
Core Concepts: OOP, Data Structures & Algorithms, DOM Manipulation, Anomaly Detection
Cloud & Data: Snowflake
IoT & Embedded: ESP32, Sensor Integration, Embedded AI, Gas & Vibration Sensors

INTERPERSONAL SKILLS

ï¿½ Communication
ï¿½ Presentation
ï¿½ Problem-Solving
ï¿½ Team Player
ï¿½ Time Management
ï¿½ Attention to Detail

ï¿½ Adaptability

WORKSHOPS & HACKATHONS

HACKXTREME'26 ï¿½ 24-Hour National-Level Hackathon, K. Ramakrishnan College of Engineering, Trichy

- ï¿½ Built an AI-powered Digital Twin model simulating urban infrastructure from IoT data for real-time prediction of traffic congestion, energy demand, and pollution levels.
- ï¿½ Recognized by Media Wave for technical performance and shortlisted for an internship opportunity.

Smart India Hackathon (SIH)

- ï¿½ Designed an automated pesticide sprinkler system using logic-based control to improve agricultural efficiency.
- ï¿½ Developed a prototype focused on reducing manual intervention while optimizing pesticide spraying.
- ï¿½ Collaborated with team members to design an innovative solution addressing real-world agricultural challenges.

MSME

- ï¿½ Presented Micro Failure Whisperer and Digital Smell as AI-driven industrial safety and predictive maintenance solutions.
- ï¿½ Demonstrated the application of machine learning and IoT technologies for early fault detection and industrial monitoring.

Snowflake Workshop ï¿½ 1 Week

- ï¿½ Completed hands-on training covering Snowflake architecture, cloud data warehousing, and SQL-based data management.
- ï¿½ Gained practical exposure to data loading, querying, and cloud-based analytics workflows.

IoT Workshop ï¿½ 1 Month

- ï¿½ Completed practical training in ESP32 programming, sensor integration, and embedded system development.
- ï¿½ Built real-time IoT applications involving sensor data acquisition and hardware-software communication.

CERTIFICATIONS

ï¿½ AI/ML (Dreams AI)
ï¿½ SQL (Oracle)
ï¿½ Java (Oracle)
ï¿½ MongoDB (ICT Academy)
ï¿½ Figma (Simplelearn)
ï¿½ Python / C & C++ (Kaggle/mygreatlearning)

EDUCATION

B.E. Computer Science and Engineering (CSE) ï¿½ PSNA College of Engineering and Technology, 2024 ï¿½ 2028 | CGPA: 8.6
Dindigul

XII ï¿½ St. John's Matriculation Higher Secondary School, Madurai May 2024 | 93%

X ï¿½ The Nallamani Matriculation Higher Secondary School, Madurai July 2022 | 92%

ACHIEVEMENTS

- ï¿½ Recognized by Media Wave at HACKXTREME'26 (24-Hour National-Level Hackathon) for technical performance and shortlisted for an internship opportunity.
- ï¿½ Solved 170+ problems on LeetCode across Arrays, Strings, Mathematics, and core DSA topics.
- ï¿½ Won mathematics competitions, demonstrating strong analytical thinking and problem-solving abilities.